

```
In[1]:= Clear["Global`*"];
```

0. Readme

1. (x', y', z') are represented by $(x3, y3, z3)$.
2. $V_{\lambda}^s, V_{\phi}^s, V_r^s, V_{\lambda\lambda}^s, V_{\lambda\phi}^s, V_{\lambda r}^s, V_{\phi\phi}^s, V_{\phi r}^s$, and V_{rr}^s in Table B1 are represented by SphericalV λ , SphericalV ϕ , SphericalVr, SphericalV $\lambda\lambda$, SphericalV $\lambda\phi$, SphericalV λr , SphericalV $\phi\phi$, SphericalV ϕr , and SphericalVrr.
3. $V_x^c, V_y^c, V_z^c, V_{xx}^c, V_{xy}^c, V_{xz}^c, V_{yy}^c, V_{yz}^c$, and V_{zz}^c in Table B2 are represented by CartesianVx, CartesianVy, CartesianVz, CartesianVxx, CartesianVxy, CartesianVxz, CartesianVyy, CartesianVyz, and CartesianVzz.
4. The integral kernels are considered and the complete expressions should be presented as $G\rho \int_{\lambda_1}^{\lambda_2} \int_{\phi_1}^{\phi_2} \int_{u_1}^{u_2} \kappa(\text{Kernel}) d u' d \phi' d \lambda'$.

1. Basic settings

```
In[2]:= A = x3 * Sin[λ] - y3 * Cos[λ];
        B = x3 * Cos[λ] + y3 * Sin[λ];

In[4]:= x = r * Cos[φ] * Cos[λ];
        y = r * Cos[φ] * Sin[λ];
        z = r * Sin[φ];

In[7]:= Δx = x - x3;
        Δy = y - y3;
        Δz = z - z3;

In[10]:= dx = -Sin[λ] * Δx + Cos[λ] * Δy;
         dy = -Sin[φ] * Cos[λ] * Δx - Sin[φ] * Sin[λ] * Δy + Cos[φ] * Δz;
         dz = Cos[φ] * Cos[λ] * Δx + Cos[φ] * Sin[λ] * Δy + Sin[φ] * Δz;

In[13]:= l = Sqrt[r^2 + x3^2 + y3^2 + z3^2 - 2 * r * (z3 * Sin[φ] + B * Cos[φ])];
```

2. GV and GGT expressions in spherical integral kernels from Table B1

```
In[14]:= SphericalVλ = -A / l^3;

In[15]:= SphericalVφ = (z3 * Cos[φ] - B * Sin[φ]) / l^3;

In[16]:= SphericalVr = -(r - z3 * Sin[φ] - B * Cos[φ]) / l^3;

In[17]:= SphericalVλλ = (-2 * r^2 + x3^2 + y3^2 - 2 * z3^2 - 3 * (x3^2 - y3^2) * Cos[2 * λ] +
                        4 * r * z3 * Sin[φ] + 4 * r * B * Cos[φ] - 6 * x3 * y3 * Sin[2 * λ]) / (2 * l^5);

In[18]:= SphericalVλφ = -3 * A * Cos[φ] * (z3 - B * Tan[φ]) / l^5;

In[19]:= SphericalVλr = 3 * A * (r - z3 * Sin[φ] - B * Cos[φ]) / l^5;

In[20]:= SphericalVφφ = (-r^2 - x3^2 - y3^2 - z3^2 + 3 * z3^2 * (Cos[φ])^2 + 2 * r * z3 * Sin[φ] +
                        2 * B * Cos[φ] * (r - 3 * z3 * Sin[φ]) + 3 * B^2 * (Sin[φ])^2) / l^5;
```

```
In[21]:= SphericalVφr = -3 * (r - z3 * Sin[φ] - B * Cos[φ]) * (z3 * Cos[φ] - B * Sin[φ]) / l^5;
In[22]:= SphericalVrr = (2 * r^2 - x3^2 - y3^2 - z3^2 -
(4 * r - 3 * z3 * Sin[φ] - 3 * B * Cos[φ]) * (z3 * Sin[φ] + B * Cos[φ])) / l^5;
```

3. GV and GGT expressions in Cartesian integral kernels from Table B2

```
In[23]:= CartesianVx = -dx / l^3;
In[24]:= CartesianVy = -dy / l^3;
In[25]:= CartesianVz = -dz / l^3;
In[26]:= CartesianVxx = (3 * dx^2 - l^2) / l^5;
In[27]:= CartesianVxy = 3 * dx * dy / l^5;
In[28]:= CartesianVxz = 3 * dx * dz / l^5;
In[29]:= CartesianVyy = (3 * dy^2 - l^2) / l^5;
In[30]:= CartesianVyz = 3 * dy * dz / l^5;
In[31]:= CartesianVzz = (3 * dz^2 - l^2) / l^5;
```

4. Differences between GV and GGT expressions in spherical and Cartesian integral kernels in Tables B1 and B2

```
In[32]:= FullSimplify[SphericalVλ - CartesianVx]
Out[32]=
0

In[33]:= FullSimplify[SphericalVφ - CartesianVy]
Out[33]=
0

In[34]:= FullSimplify[SphericalVr - CartesianVz]
Out[34]=
0

In[35]:= FullSimplify[SphericalVλλ - CartesianVxx]
Out[35]=
0

In[36]:= FullSimplify[SphericalVλφ - CartesianVxy]
Out[36]=
0

In[37]:= FullSimplify[SphericalVλr - CartesianVxz]
Out[37]=
0
```

```
In[38]:= FullSimplify[SphericalV $\phi\phi$  - CartesianVyy]
```

```
Out[38]=
```

0

```
In[39]:= FullSimplify[SphericalV $\phi r$  - CartesianVyz]
```

```
Out[39]=
```

0

```
In[40]:= FullSimplify[SphericalVrr - CartesianVzz]
```

```
Out[40]=
```

0

```
In[41]:= NotebookSave[EvaluationNotebook[]];
```